



A

PROJECT REPORT

On

**IoT based Smart Anti-theft alert and Engine lock
system for Two wheelers**

*Submitted in partial fulfilment of
the requirements for the award of the Degree of*

BACHELOR OF SCIENCE

IN

ELECTRONICS AND TELECOMMUNICATION

by

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DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION

B.J.B. AUTONOMOUS COLLEGE

BHUBANESWAR.

2025-26

CERTIFICATE

This is to certify that Mr. Pravas Nayak, Mr. Tushar Kant Pradhan and Ms. Sunayana Acharya, bearing Roll Nos 202353011, 202353010, 202353003 are the students of IMSc.ETC of BJB Autonomous College, Bhubaneswar have undergone a project work on IoT based Smart Anti-theft alert and Engine lock system for Two wheelers, at Oltron Technology from 31 January 2025 to 5 March 2026 under my supervision & guidance.

Certified that to the best of my knowledge, this work done here is very correct and accurate. During this period they have shown keen interest, dedication and hard work for the development of this project

I wish all success in their future endeavour.

Signature of the Guide with seal

Name:

Designation:



DEPT. OF ELECTRONICS AND TELECOMMUNICATION

BJB (AUTONOMOUS) COLLEGE

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BONAFIDE CERTIFICATE

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Sign. of External

Examiner

Sign. Of HOD

Dept. Of IMSc(ETC)

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No project is the outcome of the individual effort. Right from its inception to the completion of the project, We are benefited by the suggestions and valuable inputs from many people. Without their help, we would not have been able to complete this project. So, we seize this opportunity to thank all of them who have helped us to complete this project.

*We would like to express our deep sense of gratitude to our external guides **Mr. Bibhuti Kumar Behera** and **Mr. Roushan Kumar Niraj** for their inspiring guidance and cooperation for the successful completion of the project.*

*We would also like to express our hearty thanks to our internal project guide **Mr. Umakanta Samantasinghar** who has provided continuous guidance and suggestions to undertake and complete the project work in this esteemed organization.*

*We are also very much thankful to the **Head of the Department and other faculty members**, for their valuable guidance, support and cooperation to the successful completion of the project.*

Signature of the Student(s)

DECLARATION

We do hereby declare that, this project report on “IoT based Smart Anti-theft alert and Engine lock system for Two wheelers” is original and the result of our own work.

This project has not been published elsewhere or submitted to any other college/University or institute for the award of any degree or diploma. This is being submitted for the partial fulfilment of our Bachelor of Science in Electronic and Telecommunication degree.

Place:

Signature of the Student(s)

Date:

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ABSTRACT

Two-wheeler theft continues to be a significant concern, primarily due to the lack of intelligent, adaptive, and reliable security mechanisms in conventional vehicles. Most existing systems rely on basic locking mechanisms or simple alarms, which can be easily bypassed or may generate false alerts due to environmental disturbances. To address these limitations, this project proposes an IoT-based smart anti-theft alert and engine lock system that offers both pre-theft detection and active theft prevention for parked two-wheelers. In this system, when the vehicle is parked and the security mode is activated, the NodeMCU microcontroller initializes by recording the normal handle position using a joystick-based steering sensor. This initial calibration serves as a reference point for further monitoring. Any sustained deviation from this reference position—such as unauthorized turning of the handle, pushing, or rolling of the vehicle—is continuously monitored and analyzed. If the deviation exceeds a predefined threshold for a specific duration, it is identified as a potential theft attempt. This threshold-based detection mechanism is specifically designed to reduce false triggering caused by minor vibrations, accidental touches, or environmental factors such as wind. To further enhance reliability, a vibration and tilt sensor is incorporated as a secondary confirmation mechanism. This sensor is activated only when significant physical disturbances are detected, ensuring that the system responds only to genuine tampering attempts rather than negligible movements. By combining both steering-based detection and vibration analysis, the system achieves a higher level of accuracy and robustness. Once a theft attempt is confirmed for a few seconds, the system initiates immediate countermeasures. A relay-based engine immobilization mechanism is activated to disable the vehicle's ignition system, which is demonstrated in this project using a DC motor setup. This effectively prevents the vehicle from being started or moved further. At the same time, audible alerts (buzzer) and visual indicators (LCD display) are triggered to warn nearby individuals and deter the thief. In addition to local alerts, the system leverages IoT connectivity to send real-time notifications to the vehicle owner. This allows remote monitoring and ensures that the user is instantly informed about any suspicious activity, even when they are not near the vehicle. The notification system enhances responsiveness and enables timely action. The engine remains locked until the system is manually disarmed by an authorized user, ensuring complete control over vehicle access. Overall, the proposed system is reliable, scalable, and cost-effective. It integrates key concepts of embedded systems, sensor-based detection, and IoT communication, making it highly suitable for academic implementation as well as practical real-world vehicle security applications.

INTRODUCTION

2.1 Problem Statement:-

Most conventional anti-theft systems suffer from two major flaws:

1. **High False Alarm Rates:** Systems relying solely on vibration sensors often trigger due to passing heavy vehicles or environmental factors, leading owners to eventually ignore the alerts.
2. **Lack of Real-Time Communication:** A standard alarm is only useful if the owner is within earshot. If the vehicle is stolen while the owner is away, there is no way to track the event or intervene remotely.

2.2 Proposed Solution:-

This project introduces an **IoT-based Smart Anti-Theft Alert and Engine Lock System**. By leveraging the NodeMCU (ESP8266) microcontroller, the system creates a digital perimeter around the vehicle's physical state. Unlike traditional alarms, this system uses a joystick-based steering sensor to monitor handle orientation, ensuring that alerts are only triggered by actual movement rather than mere vibrations.

2.3 System Analysis:-

2.3.1 Existing System:-

The existing security mechanisms for two-wheelers primarily rely on mechanical locks and basic acoustic alarms. These systems generally consist of:

- **Mechanical Disc/Handle Locks:** Physical barriers that can be bypassed using master keys or manual force.
- **Passive Vibration Alarms:** Simple electronic modules that trigger a loud siren when the vehicle is shaken.

2.3.2 Drawbacks of the Existing System:

- **High False Alarm Rate:** Standard vibration sensors cannot distinguish between a thief and environmental factors like heavy rain, thunder, or a passing heavy vehicle.
- **Acoustic Limitation:** The alarm is only effective if the owner or a bystander is within earshot. In a noisy city or remote parking, the alarm goes unnoticed.
- **No Active Prevention:** Most existing systems only provide an alert; they do not physically prevent the thief from starting the engine once the ignition is hot-wired.
- **Lack of Tracking:** Once a vehicle is stolen, the existing system provides no way to track its location, making recovery nearly impossible.

2.4 Proposed System:-

The proposed system overcomes these limitations by introducing Intelligence and Connectivity into the security loop. Unlike the existing system, this prototype utilizes:

- **Active Monitoring:** Using a Joystick-based steering sensor to detect the specific physical movement of the handlebars rather than just vibrations.
- **IoT Connectivity:** Through the NodeMCU, the system sends an instant notification to a web dashboard, removing the "earshot" limitation.
- **Engine Immobilization:** A Relay-based engine lock that physically disconnects the ignition circuit, preventing the vehicle from being driven away even if the mechanical lock is broken.
- **Real-Time Recovery:** Integrated GPS tracking allows for the precise location of the vehicle to be monitored on a global map.

2.5 Objectives:-

- **Smart Detection:** To create a system that knows the difference between someone bumping into a bike and someone actually trying to steal it. By using a Joystick sensor, the project aims to detect if the handle is being turned or pushed, which is a much more reliable sign of theft than just simple vibrations.
- **Automatic Engine Lock:** To build a "kill-switch" using a Relay. The goal is to physically stop the engine (simulated by a motor) from starting as soon as a theft is detected. This moves the system from just "making noise" to actually stopping the thief in their tracks.
- **Stay Connected Anywhere:** To use the NodeMCUs Wi-Fi to send an instant alert to the owner's phone. The objective is to make sure distance doesn't matter; whether the owner is 10 meters or 10 miles away, they get a notification the second their bike is touched.
- **Stop False Alarms:** To combine two sensors (Joystick + Vibration) so the alarm doesn't go off for no reason. This "double-check" system ensures the owner only gets an alert when there is a real threat, preventing them from getting annoyed by fake alarms.
- **Live Tracking & Updates:** To use a GPS module to find the bike's exact location and an LCD screen to show the system status. This ensures the owner can track the vehicle on a map and see if the system is "Armed" or "Locked" just by looking at the display.

PROJECT OVERVIEW AND METHODOLOGY

3.1 Project Overview:-

In the modern era, two-wheelers have become one of the primary modes of transportation for millions of people due to their affordability, fuel efficiency, and ability to navigate through heavy traffic. They are especially popular among students, office workers, and delivery personnel in urban as well as semi-urban areas. However, their portability and the relatively simple mechanical structures used in many bikes and scooters make them particularly vulnerable to theft. Unlike larger vehicles, two-wheelers can often be moved or lifted with minimal effort, making them easy targets for thieves. In addition, the relative ease of bypassing mechanical locks and ignition systems further increases the risk of unauthorized access. Traditional security measures, such as physical handle locks, wheel locks, and basic alarm systems, provide only a limited level of protection. While these methods may deter casual thieves, they are often ineffective against experienced criminals who possess the tools and knowledge to break locks or manipulate ignition wiring within a short time. In many cases, stolen two-wheelers are quickly transported or dismantled for parts before the owner even realizes the vehicle is missing. As a result, relying solely on passive security measures is no longer sufficient in today's environment. To address these vulnerabilities, there is a growing need for smart and active security systems that combine modern electronics, sensors, and communication technologies. Such systems can continuously monitor the condition of the vehicle and immediately notify the owner if suspicious activity is detected. More importantly, they can also take preventive actions, such as disabling the ignition system or activating alerts, to physically prevent the vehicle from being operated. By integrating technologies like IoT connectivity, real-time monitoring, and automated control mechanisms, modern anti-theft systems can significantly enhance vehicle security and provide greater peace of mind to owners.

3.2 System Design & Architecture:-

The project manages the operation through four simple stages:

- **Input Stage:** The input stage utilizes digital interrupt logic. The joystick sensor is calibrated such that any movement beyond a fixed physical angle triggers a digital signal to the NodeMCU, ensuring instantaneous detection of handle tampering without the latency of continuous analog processing.
- **Processing Stage:** The NodeMCU compares live data from the sensors against that baseline. It looks for any big changes like the handle being turned or the bike being shaken.
- **Output Stage:** If the system detects a theft, it immediately triggers the Relay to lock the engine, sounds the Buzzer to alert people nearby, and updates the LCD screen to show a "Mishandle or Vibration" message.
- **Communication Stage:** At the same time, the GPS module grabs the vehicle's exact location and sends it to the owner's phone via Wi-Fi.

This section describes how the different components of the system interact and work together to provide effective protection for the vehicle. The navigation functionality is achieved using a GPS module, which communicates with the NodeMCU through a simple serial interface using TX and RX pins. This module continuously provides accurate geographical coordinates of the vehicle, enabling real-time tracking and allowing the owner to monitor the vehicle's location remotely through IoT connectivity. For user interaction and system feedback, a 16×2 LCD display is used as a simple dashboard interface. It provides real-time status updates by displaying messages such as “Mishandle” or “Vibration,” depending on the type of disturbance detected. This ensures that anyone near the vehicle can immediately understand the current condition of the system, enhancing transparency and usability. Power management is handled by a dedicated Power Supply Unit (PSU), which ensures that all components receive appropriate and stable voltage levels according to their requirements. High-power components such as the relay module and DC motor require a relatively higher voltage and current, whereas the NodeMCU operates at a lower voltage level. The PSU effectively distributes power to each component, maintaining system stability, preventing voltage fluctuations, and ensuring safe operation of the entire circuit. The engine control mechanism is implemented using a relay configured in a Normally Closed (NC) arrangement. Under normal conditions, the relay allows the circuit to remain closed, enabling the engine to operate without interruption. However, when a theft attempt is detected and confirmed by the system, the NodeMCU sends a signal to trigger the relay. This causes the relay to switch states and open the circuit, thereby cutting off power to the engine and effectively immobilizing the vehicle. This immediate interruption ensures that the vehicle cannot be started or moved, providing a strong layer of protection against theft.

HARDWARE AND SOFTWARE REQUIREMENTS

4.1 Hardware Requirements:-

Slno.	Name	Specification	Quantity	Purpose
1	NodeMCU	ESP8266	1	Used as the main microcontroller with built-in Wi-Fi capability.
2	Joystick Sensor	PS2 Joystick Module	1	Used as the steering/handle movement detection sensor.
3	Vibration/Tilt Sensor	SW520D	1	Used as the vibration/tilt detection sensor module.
4	GPS Module	NEO-6M	1	Used as the GPS module for location tracking.
5	16×2 LCD with I2C Adapter	PCF8574	1	Used for displaying system status with simplified I2C communication.
6	5V Relay Module	---	1	Used to control and cut off the engine circuit.
7	Piezo Buzzer Module	---	1	Used for generating the alarm sound.
8	DC Motor	---	1	Used to simulate the vehicle engine in the prototype.

9	Power Supply Unit	---	1	Used to provide stable voltage to the system components.
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4.2 Software Requirements:-

Slno	Name	Purpose
1	Arduino IDE	Used to write, compile, and upload the C++ program to the NodeMCU microcontroller.
2	ESP8266 Board Library	Enables the Arduino IDE to program and communicate with the ESP8266-based NodeMCU board.
3	LiquidCrystal_I2C Library	Controls the 16×2 LCD display through I2C communication to show system status messages.
4	ESP8266WebServer Library	Allows the NodeMCU to host a web server for remotely monitoring the system through a browser.
5	Wire Library	Handles I2C communication between the NodeMCU and connected devices like the LCD module.

COMPONENT STUDY

5.1 NodeMCU:-

The NodeMCU is an open-source, LUA-based firmware and development board specifically designed for Internet of Things (IoT) applications. At its core, it features the ESP8266EX System-on-Chip (SoC), which integrates a 32-bit Tensilica Xtensa LX106 microcontroller and a standard IEEE 802.11 b/g/n Wi-Fi transceiver. This integration allows the module to handle complex computational tasks while simultaneously maintaining a persistent wireless connection to the local network. It operates at 3.3 V (or 5 V through Micro-USB). It supports a clock speed of 80 MHz to 160 MHz and includes 4 MB of flash memory. The module supports communication interfaces such as SPI, I2C, UART, ADC, and PWM.

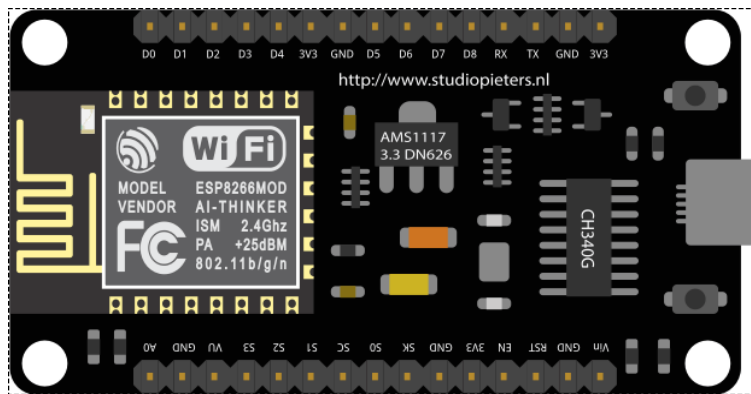


Fig 5.1.1:- NodeMCU

Multiple GPIO pins on the board allow us to connect the board with other peripherals and are capable of generating PWM, I2C, SPI, and UART serial communications. The interface of the module is mainly divided into two parts including both Firmware and Hardware where former runs on the ESP8266 Wi-Fi SoC and later is based on the ESP-12 module. The firmware is based on LUA, a scripting language that is easy to learn, giving a simple programming environment layered with a fast scripting language that connects you with a well-known developer community. Open source firmware gives you the flexibility to edit, modify and rebuilt the existing module and keep changing the entire interface until you succeed in optimizing the module as per your requirements. USB to UART converter is added on the module that helps in converting USB data to UART data which mainly understands the language of serial communication. Instead of the regular USB port, MicroUSB port is included in the module that connects it with the computer for dual purposes:- programming and powering up the board. The board incorporates status LED that blinks and turns off immediately, giving you the current status of the module if it is running properly when connected with the computer. The ability of module to establish a flawless WiFi connection between two channels makes it an ideal choice for incorporating it with other embedded devices like Raspberry Pi.

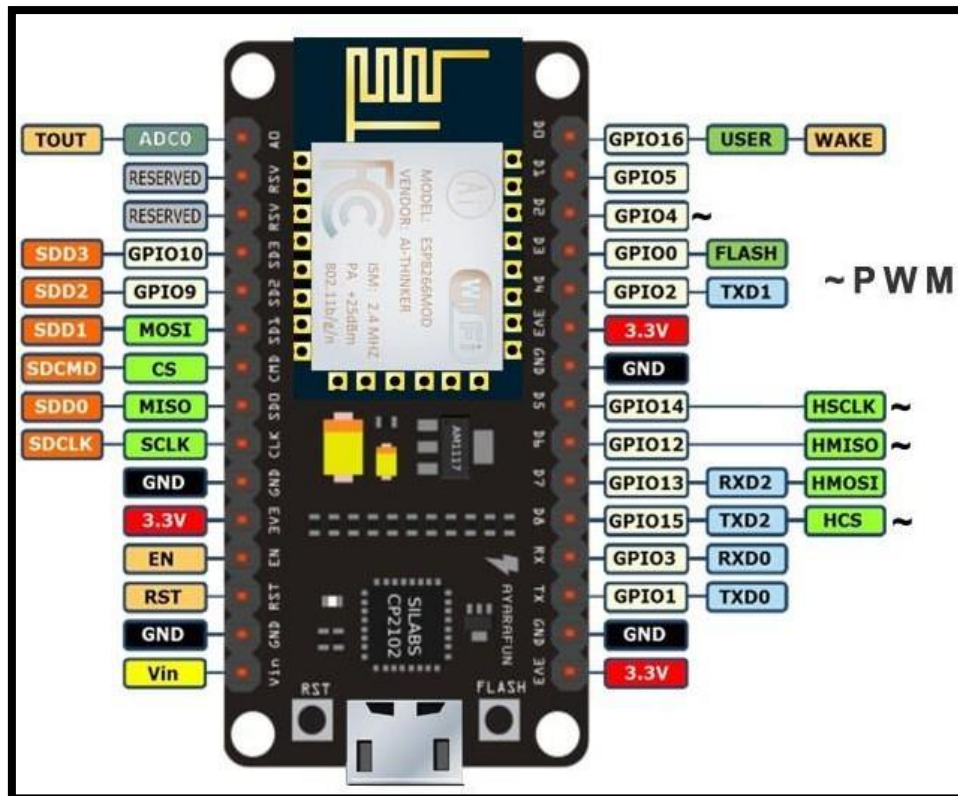


Fig 5.1.2:- NodeMCU Pin layout

The NodeMCU is an open-source, low-cost development board that features the ESP8266 chip with integrated Wi-Fi, making it a powerful yet affordable choice for IoT applications. It offers Arduino-like hardware functionality with an interactive and programmable environment, including a MicroUSB port for power, a USB-to-UART converter for seamless communication, and dedicated Reset and Flash buttons. Equipped with a versatile set of GPIO pins and a status LED, it supports an advanced hardware API that significantly reduces redundant configuration tasks. While it can be coded exactly like an Arduino using the IDE, it also supports interactive Lua scripting, providing a flexible and efficient platform for manipulating hardware. In this project, the NodeMCU acts as the central processing unit (CPU). It is responsible for monitoring the digital state of the joystick and vibration/tilt sensors, executing the anti-theft logic, and triggering the relay-based engine lock. Furthermore, it hosts the localized ESP8266WebServer, which generates the real-time security dashboard accessible via mobile browsers. By utilizing its General Purpose Input/Output (GPIO) pins, the NodeMCU ensures seamless communication between the hardware sensors and the digital cloud interface, providing a scalable and low-power solution for vehicle security.

5.2 Sensors:-

A sensor, as the name suggests, is used to sense any type of disturbances. A sensor can give analog as well as digital value. A sensor (also called detector) is a device that measures a physical quantity and converts it into a signal which can be read by an observer or by an instrument. A sensor is a device which receives and responds to a signal. A sensor's sensitivity indicates how much the sensor's output changes when the measured quantity changes. Sensors that measure very small changes must have very high sensitivities. Sensors also have an impact on what they measure. The sensors that are used in our project are:-

5.2.1 Joystick Sensor (PS2 Joystick Module):- A joystick sensor is an input device used to detect movement in different directions. It is commonly used in gaming controllers, robotics, and embedded systems to control the direction of movement of a device. In IoT and microcontroller-based projects, a joystick sensor allows the user to provide manual input by moving the stick along the X-axis and Y-axis. The joystick module usually consists of two potentiometers that measure movement in horizontal and vertical directions. The joystick sensor module operates at a voltage range of 3.3 V to 5 V and uses a dual-axis potentiometer to detect movement. It supports digital and analog interfacing (with pin D7 utilized) and can detect movement along the X, Y, and Z axes. The module is made of ABS plastic and metal, with approximate dimensions of $4.0 \times 2.6 \times 3.2$ cm.

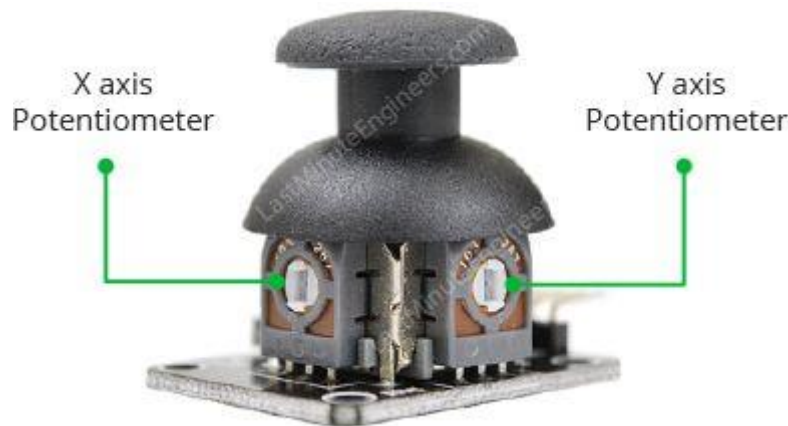


Fig 5.2.1-1:- Joystick Sensor

When the joystick is moved, the resistance of the potentiometers changes, producing different voltage outputs. These voltage values are read by the microcontroller through analog input pins, which allows the system to determine the position and direction of the joystick. Most joystick modules also include a push button switch that gets activated when the joystick is pressed down. This feature provides an additional digital input that can be used to trigger specific actions in the system.

In this system, the joystick is interfaced with the NodeMCU via Pin D7. While the sensor is capable of high-resolution analog output, it is configured here with a digital logic threshold to ensure a rapid response time. It acts as the primary "pre-theft" detection mechanism by monitoring the physical orientation of the vehicle's handle. Unlike passive sensors that wait for a shock, this active sensor detects the specific angular motion involved in steering or pushing the bike. When the handle is moved beyond a predefined physical limit, the sensor pulls the input pin to a Logic LOW state, triggering the NodeMCU to immediately initiate the security protocol and engine immobilization.

5.2.2 Vibration/Tilt Sensor (SW520D):- The SW-520D is a specialized ball-actuated kinetic sensor designed to detect changes in physical orientation, inclination, and mechanical disturbances. Structurally, the sensor consists of a small, hermetically sealed gold-plated brass cylinder containing two high-precision metallic balls. This internal configuration allows the device to function as a non-directional rolling-ball tilt switch. The vibration/tilt sensor operates within a voltage range of 3.3 V to 5 V and uses a ball-actuated mechanical switching mechanism to detect motion or tilt. It has a contact resistance of less than 10 ohms and provides a digital output interface (with pin D6 utilized). The sensor is enclosed in a gold-

plated brass cylindrical housing and can detect tilt or vibration when the angle exceeds 15 degrees.

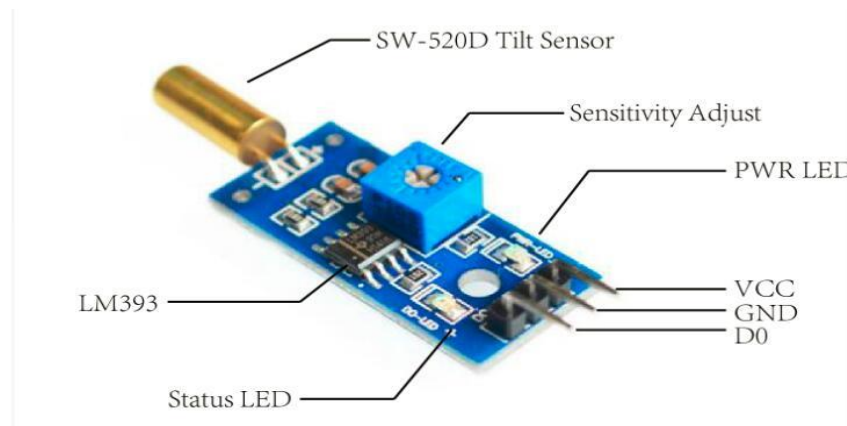


Fig 5.2.2-1:- Vibration/Tilt Sensor

It is widely utilized in electronics for its durability and its ability to provide a clean digital trigger in response to physical movement. The operational principle of the SW-520D is based on mechanical contact logic. When the sensor is positioned at a specific angle (typically upright), the internal metallic balls roll to one end of the gold-plated cylinder, bridging the conductive pins and completing an electrical circuit. This state represents a closed connection with minimal contact resistance. Conversely, when the sensor is tilted beyond a critical threshold usually between 15° and 90° or subjected to significant centrifugal force or vibration, the balls roll away from the terminal points. This physical displacement breaks the electrical continuity, resulting in an "open circuit" state. Unlike mercury-based tilt switches, the SW-520D is an environmentally friendly, solid-state alternative that offers high sensitivity without the hazards of toxic chemicals. The gold plating on the internal housing and terminal pins ensures excellent conductivity and prevents oxidation over time, which is critical for maintaining signal integrity in long-term deployments. Because it provides a binary (on/off) output, it is highly compatible with microcontrollers, requiring no complex analog-to-digital conversion to interpret orientation data.

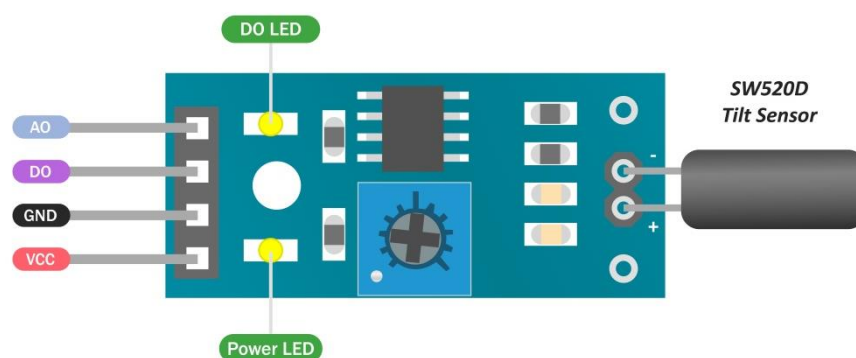


Fig 5.2.2-2:- Vibration/Tilt Sensor Pin Layout

In this security system, the sensor is interfaced with Pin D6 of the NodeMCU. While the joystick monitors steering rotation, this module acts as a "Tamper Detection" layer. It is specifically utilized to identify if a parked bike is being lifted off its side-stand or if the

chassis is being shaken or actions typical of a forced ignition or a theft attempt. By using the SW-520D, the system achieves a higher degree of reliability compared to standard vibration springs, as it is less prone to false triggers from environmental noise like heavy wind or passing traffic. When a tilt or vibration event is detected, the sensor sends a Logic LOW signal to the microcontroller, triggering the engine lock and updating the web dashboard immediately.

5.3 Buzzer:-

A buzzer is an audio signalling device that serves as the primary acoustic interface in electronic systems. The most common type used in microcontroller projects is the Active Piezoelectric Buzzer, which contains an internal oscillation circuit. When a DC voltage is applied, a ceramic piezoelectric element vibrates at a specific frequency, creating a sharp, audible sound. The active buzzer operates within a voltage range of 3.3 V to 5 V and has a resonant frequency of 2300 ± 300 Hz. It produces a sound pressure level of at least 85 dB at a distance of 10 cm while consuming less than 30 mA of current. This buzzer is an active type with an internal oscillator, and its compact dimensions are approximately 12 mm $\times$ 9.5 mm.



Fig 5.3.1:- Buzzer

This design is highly efficient, requiring very little current to produce a high-decibel output. In the context of vehicle security, the buzzer acts as the "Local Alarm" system. It is interfaced with the NodeMCU to provide immediate audible alerts during a security breach. Unlike the web dashboard, which provides remote monitoring, the buzzer ensures that any unauthorized person near the vehicle is instantly deterred by a loud warning tone. In the system logic, when either the tilt sensor or the joystick triggers a theft alert, the NodeMCU sends a pulse-width modulated (PWM) or a high-logic signal to the buzzer's driver circuit. This generates a continuous or pulsing alarm sound that persists until the system is reset or disarmed. Beyond alerts, the buzzer is also used to provide "system-ready" chirps, giving the user confirmation that the anti-theft protocols have been successfully armed.

5.4 Global Positioning System (GPS):-

The NEO-6M GPS Module is a high-performance, standalone Global Positioning System (GPS) receiver module designed for high-precision location tracking and navigation. It is built upon the u-blox 6 positioning engine, which features 50 channels and a sophisticated architecture that allows it to track multiple satellite constellations simultaneously. The module is engineered to operate under low-power conditions while maintaining a high sensitivity of -161 dBm, making it capable of acquiring signals even in challenging urban

environments where line-of-sight to the sky may be partially obstructed. The GPS module operates with a supply voltage ranging from 3.0 V to 5.0 V and communicates with the microcontroller using the UART (serial) communication protocol with a default baud rate of 9600. It provides a navigation update rate between 1 Hz and 5 Hz and offers a horizontal position accuracy of about 2.5 meters. The module can support a maximum velocity of 500 m/s and functions reliably within an operating temperature range of -40°C to +85°C.

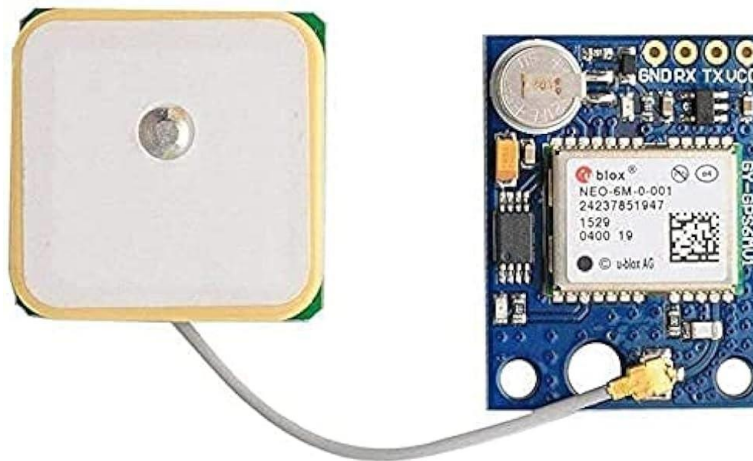


Fig 5.4.1:- NEO-6M GPS Module with EEPROM

One of the defining technical features of this specific version is the integration of an onboard EEPROM (Electrically Erasable Programmable Read-Only Memory). Unlike standard modules that lose their configuration upon power-off, the EEPROM allows the NEO-6M to store user-defined settings, such as baud rate and update frequency. More importantly, it stores Ephemeris and Almanac data and detailed information about the orbital paths of GPS satellites. By retaining this data in non-volatile memory, the module can perform a "Warm Start" or "Hot Start," significantly reducing the Time To First Fix (TTFF) from approximately 27 seconds (Cold Start) to under 1 second. The module communicates using the National Marine Electronics Association (NMEA) protocol via a standard UART serial interface. It outputs a series of standardized data strings, known as NMEA sentences, which contain comprehensive information including latitude, longitude, altitude, velocity, and Coordinated Universal Time (UTC). To ensure stable signal reception, the module is typically paired with a high-gain ceramic patch antenna. Additionally, the inclusion of a rechargeable backup battery ensures that the onboard Real-Time Clock (RTC) remains active during power interruptions, further accelerating the synchronization process once power is restored. The GPS module serves as the primary tracking engine for the anti-theft system. While the Tilt and Joystick sensors detect the start of a theft, the NEO-6M ensures the vehicle remains traceable if it is successfully moved. It communicates with the NodeMCU via UART (Universal Asynchronous Receiver-Transmitter) serial communication. In the event of an alert, the NodeMCU extracts the raw NMEA data sentences from the GPS, parses them into human-readable coordinates, and pushes this data to the real-time web dashboard. This allows the vehicle owner to monitor the bike's location in real-time from any remote location via a mobile browser.

5.5 16x2 Liquid Crystal Display (LCD):-

A 16x2 Liquid Crystal Display (LCD) is a basic alphanumeric display module widely used in embedded systems to provide visual feedback. The LCD display module operates at an

operating voltage of 4.7 V to 5.3 V and typically consumes 120 mA to 160 mA of current when the backlight is on. It has character dimensions of approximately 2.95 mm × 5.55 mm and communicates with the microcontroller using the I2C serial communication protocol. The common I2C addresses are 0x27 or 0x3F, and the display can operate within a temperature range of −20°C to +70°C. The "16x2" designation indicates that the screen can display two rows of text, with each row accommodating up to 16 characters. Each character is rendered within a 5 × 8 pixel matrix. The display typically utilizes the HD44780 controller, which manages the complex task of controlling the liquid crystal segments to form recognizable letters, numbers, and symbols.

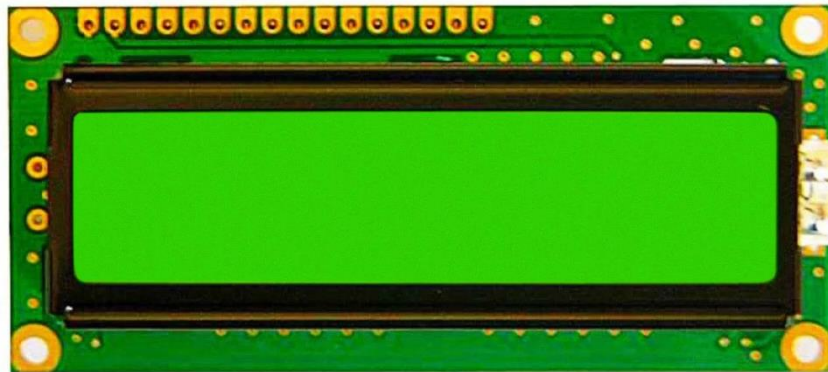


Fig 5.5.1:- 16x2 LCD Display

The standard 16x2 LCD requires at least 6 to 10 General Purpose Input/Output (GPIO) pins for data and control signals (in 4-bit or 8-bit mode). To optimize pin usage on microcontrollers like the NodeMCU, an I2C (Inter-Integrated Circuit) Adapter is integrated into the module. This adapter uses the PCF8574 I/O expander chip to convert the 16-pin parallel interface into a simple 2-wire serial interface. This reduces the required connection points to just two data lines: SDA (Serial Data) and SCL (Serial Clock). The display works on the principle of light modulation. It consists of a layer of liquid crystal molecules sandwiched between two transparent electrodes and two polarizing filters. When an electrical field is applied to specific segments, the molecules align in a way that blocks or allows light to pass through from the LED backlight, creating visible characters on the screen. The I2C adapter also includes a potentiometer (a variable resistor) to manually adjust the contrast of the screen, ensuring that the characters are visible under different lighting conditions.

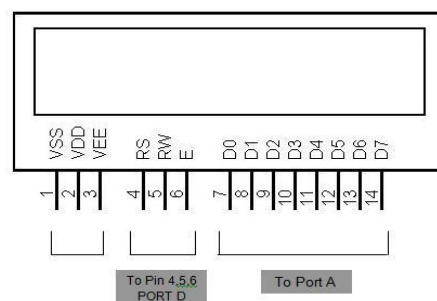


Fig 5.5.2:- LCD Display Pin Layout

In our project, the 16x2 LCD serves as the primary Local User Interface (UI), providing immediate visual feedback on the system's operational status. While the IoT dashboard handles remote monitoring, the LCD allows for on-site diagnostics by displaying real-time

messages such as "Mishandle", "Normal" or "Vibration" as well as the IP address of the connected device. By utilizing the I2C protocol, the display is able to communicate with the NodeMCU using only two data pins, which optimizes the hardware layout and ensures that more GPIO pins remain available for the security sensors and the engine-lock relay.

5.6 Relay:-

A Relay is an electromechanical switch that utilizes electromagnetism to control a high-power circuit using a low-power signal. It acts as a bridge between a microcontroller (like the NodeMCU) and heavy-duty loads such as motors, lights, or ignition systems. The relay module operates with a 5 V DC coil voltage and requires a trigger current of about 15 mA to 20 mA. It can switch loads up to 250 V AC or 30 V DC with a maximum switching current of 10 A. The relay uses an SPDT (Single Pole Double Throw) contact type and has a switching time of less than 10 ms. The core components include an electromagnetic coil, an armature, and a set of contacts. When current flows through the coil, it creates a magnetic field that pulls the armature, causing the physical contacts to either "make" or "break" the electrical connection. The 5V Relay module is specifically designed for compatibility with digital logic.



Fig 5.6.1:- Relay

It features an integrated Optocoupler (or transistor driver) that provides Galvanic Isolation. This is a critical safety feature that ensures the high-voltage/high-current load circuit is physically and electrically separated from the sensitive 3.3V or 5V logic of the microcontroller, preventing back-EMF (electromotive force) from damaging the processor. The module typically offers three output terminals: Common (COM), Normally Open (NO), and Normally Closed (NC). In electronic switching applications, the relay is favored for its ability to handle both AC and DC loads with high efficiency. Most modules include a status LED to indicate when the coil is energized and a "flyback" diode (1N4007 or similar) connected across the coil to suppress voltage spikes during switching. Because it is a mechanical switch, it provides a very low contact resistance when closed and an infinite resistance when open, making it an ideal choice for complete circuit interruption or "immobilization" tasks.

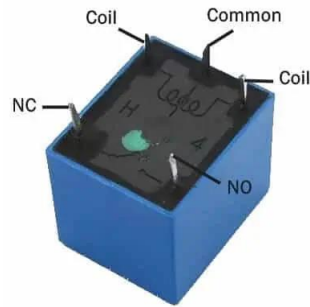


Fig 5.6.2:- Relay Pins

In this project, the 5V Relay Module functions as an Engine Immobilizer. It is wired in a Normally Closed (NC) configuration, allowing the vehicle's motor to run under standard conditions. When a security breach is detected, the NodeMCU sends a signal to the relay, which energizes an internal coil and physically breaks the ignition circuit. This provides Galvanic Isolation, allowing the low-power microcontroller to safely disconnect the high-current engine power without electrical damage, effectively preventing unauthorized use of the vehicle.

5.7 DC Motor:-

A Direct Current (DC) Motor is an electrical machine that converts direct current electrical energy into mechanical energy. The DC motor operates within a voltage range of 3 V to 12 V DC and typically provides a no-load speed between 1000 and 15000 RPM. It has a standard shaft diameter of 2.0 mm and supports bi-directional rotation (clockwise and counter-clockwise).



Fig 5.7.1:- DC Motor

The motor uses a brushed commutation mechanism for its operation. The motor operates on the principle of Lorentz Force, which states that when a current-carrying conductor is placed in a magnetic field, it experiences a mechanical force. In a standard brushed DC motor, this force creates a torque that rotates the internal shaft, providing the mechanical motion required for various applications. The internal construction typically consists of a Stator (permanent magnets that provide a constant magnetic field) and a Rotor or armature (the rotating part containing coils). When power is applied, the interaction between the magnetic field of the stator and the magnetic field generated by the rotor coils causes the shaft to spin. The speed of the motor is directly proportional to the supplied voltage, while the torque is proportional to the current. Because these motors can be easily controlled via microcontrollers using relays or motor drivers, they are the standard choice for simulating mechanical movement in electronic prototypes.

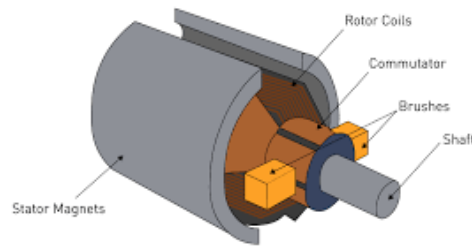


Fig 5.7.2:- DC Motor Internal Diagram

In this anti-theft system, the DC Motor represents the Vehicle's Engine or Ignition System. It serves as the primary load controlled by the relay module. Under normal "Armed" conditions, the motor remains powered, simulating a functional engine. However, the moment a security breach is detected, the relay cuts the power supply to the motor, simulating an automatic engine kill-switch. This provides a clear, physical demonstration of the system's ability to immobilize the vehicle in real-time.

5.8 Power Supply Unit:-

The Power Supply Unit is the fundamental block of any electronic system, responsible for providing stable and regulated voltage to all integrated components. Since different modules in an embedded system often have varying voltage requirements such as 3.3V for the NodeMCU logic and 5V for the Relay and LCD. The PSU must ensure that these levels are maintained despite fluctuations in the input source. A reliable power supply prevents erratic behavior, sensor inaccuracies, and permanent hardware damage caused by overvoltage.

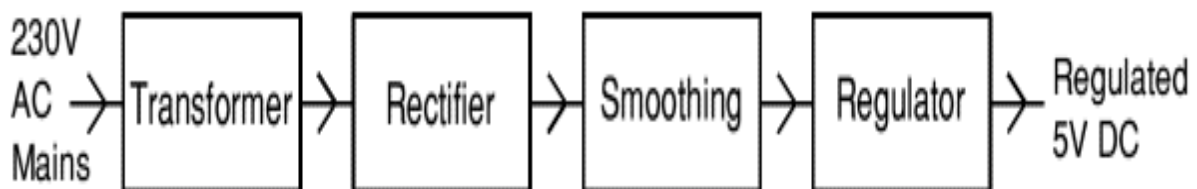


Fig 5.8.1:- Block Diagram of the Power Supply Unit

Power supply is the circuit from which we get a desired dc voltage to run the other circuits. The voltage we get from the main line is 230V AC but the other components of our circuit require 5V DC. Hence a step-down transformer is used to get 12V AC which is later converted to 12V DC using a rectifier. The output of rectifier still contains some ripples even though it is a DC signal due to which it is called as Pulsating DC. To remove the ripples and obtain smoothed DC power filter circuits are used. Here a capacitor is used. The 12V DC is rated down to 5V using a positive voltage regulator chip 7805. Thus a fixed DC voltage of 5V is obtained.

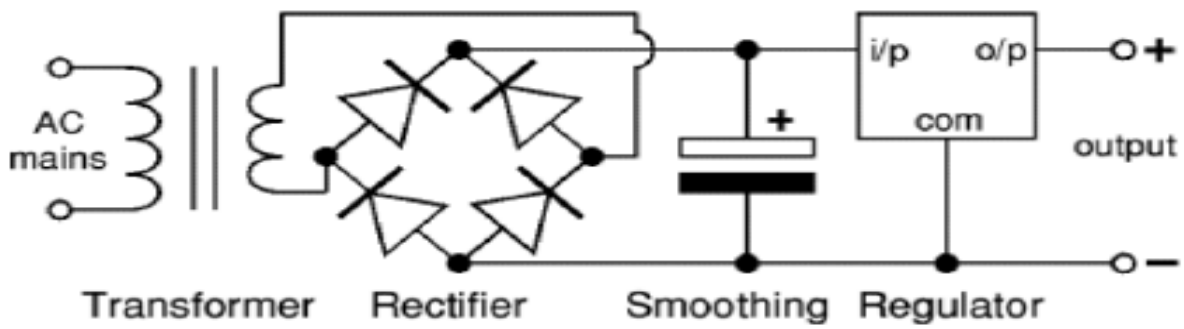


Fig 5.8.2:- Circuit Diagram of the Power Supply Unit

Each of the blocks is described in more detail below:

- Transformer - steps down high voltage AC mains to low voltage AC.
- Rectifier - converts AC to DC, but the DC output is varying.
- Capacitor - smoothes the DC from varying greatly to a small ripple.
- Regulator - eliminates ripple by setting DC output to a fixed voltage.

The Power Supply Unit in this project serves as the "heart" of the hardware, distributing energy to the NodeMCU, the security sensors, and the output peripherals. It is designed to handle the high-current draw of the DC Motor and Relay while simultaneously providing the clean, low-current power required for the GPS and IoT communication. By centralizing the power distribution, the PSU ensures that the system remains operational for long periods, providing consistent protection and real-time data transmission to the web dashboard.

BLOCK DIAGRAM

The working of the IoT-based anti-theft system as shown in the block diagram begins when the power supply is turned on and the NodeMCU microcontroller initializes all connected components. The microcontroller starts monitoring the sensors and communication modules continuously while displaying system status information on the LCD display. Under normal conditions, the vibration/tilt sensor and joystick remain inactive, and the digital input pins of the NodeMCU stay at a HIGH logic level due to the internal pull-up configuration. In this state, the relay remains in its default position and the motor or ignition circuit remains connected through the Normally Closed terminal, allowing the vehicle to operate normally. When an unauthorized person attempts to move or tilt the vehicle, the vibration/tilt sensor detects the disturbance and momentarily closes its internal contact. This action connects the microcontroller input pin to ground, changing the voltage level from HIGH to LOW. The NodeMCU immediately detects this change and interprets it as a possible theft attempt.

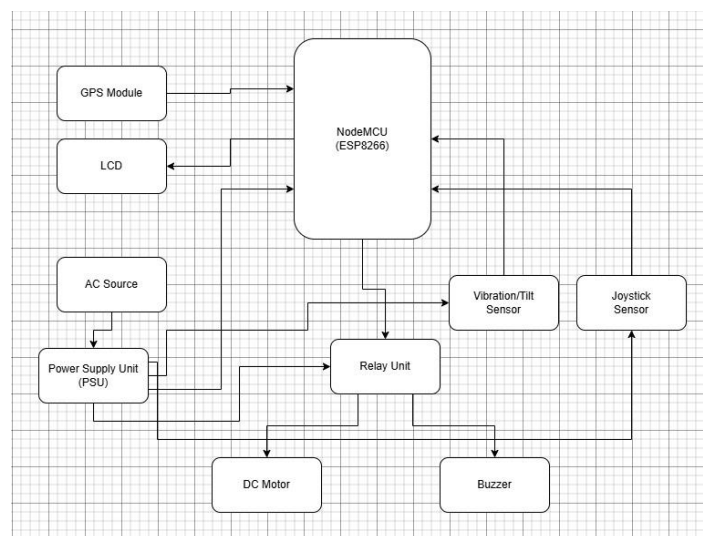


Fig 6.1:- Block Diagram of the Circuit

Once the theft condition is detected, the microcontroller executes the programmed security response. It sends a control signal to the relay module, which energizes the relay coil and changes the internal switch position. This action opens the normally closed contact and interrupts the electrical connection to the motor or ignition circuit. As a result, the engine stops functioning, preventing the vehicle from being driven away. At the same time, the NodeMCU reads location data from the GPS module through the UART communication interface. The GPS module provides accurate location coordinates in the form of NMEA sentences, which contain latitude, longitude, and time information. The microcontroller processes this data and can use it to determine the exact location of the vehicle. The system can also display important information such as system status, alert messages, or location data on the LCD screen using the I2C interface. Because the I2C protocol requires only two communication lines, it simplifies the wiring while still allowing the display to show real-time information. Through this sequence of operations, the system continuously monitors the vehicle for unauthorized movement, detects suspicious activity using sensors, disables the engine using the relay, and obtains location information using the GPS module. This integrated operation ensures that the vehicle remains protected from theft while also allowing tracking capabilities for enhanced security.

CIRCUIT ANALYSIS

The circuit of the IoT-based anti-theft alert system is built around the NodeMCU ESP8266, which acts as the central control unit of the system. It monitors sensor inputs, processes signals, and controls output devices such as the relay and LCD display. Due to its built-in Wi-Fi capability and processing power, the NodeMCU is widely used in IoT applications for monitoring and automation. A stable power supply is required for proper circuit operation. The supply begins with a step-down transformer that reduces the AC mains voltage to a lower level. This AC voltage is converted into pulsating DC using a bridge rectifier. A filter capacitor is then used to smooth the output voltage by reducing ripples. The filtered voltage is regulated using the LM7805 voltage regulator, which provides a constant 5V output. This regulated voltage powers the relay module and LCD display, while the NodeMCU internally operates at 3.3V. Additional decoupling capacitors help reduce noise and stabilize the voltage. The detection system uses motion sensing devices such as the SW-520D vibration sensor and a joystick module connected to the digital input pins of the NodeMCU. These pins use internal pull-up resistors, so the default state remains HIGH. When the vehicle is moved or tilted, the vibration sensor momentarily connects the input pin to ground, changing the voltage from HIGH to LOW. The microcontroller detects this change and identifies it as a possible theft attempt.

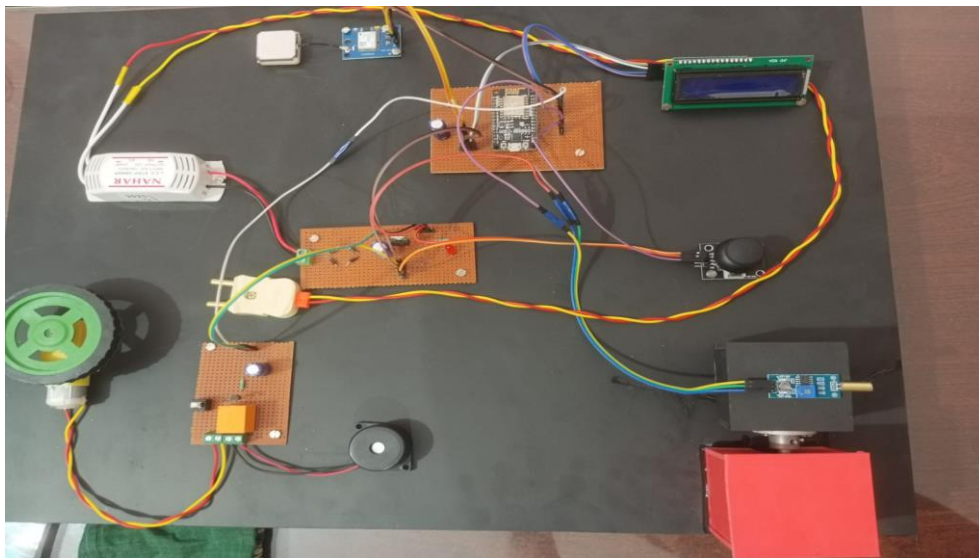


Fig 7.1:- Hardware setup of IoT based smart anti-theft alert and engine lock system

A 5V electromagnetic relay is used to control the motor or ignition circuit. The motor is connected to the Normally Closed (NC) terminal, allowing normal operation when no disturbance is detected. When theft activity is sensed, the NodeMCU activates the relay coil, opening the NC contact and stopping the motor. The relay also provides galvanic isolation between the microcontroller and the high-current motor circuit. For communication, the I2C protocol is used to interface the 16x2 LCD display with the NodeMCU using only two lines (SDA and SCL). The UART protocol is used to communicate with the NEO-6M GPS module, which receives satellite signals and provides location data in the form of NMEA sentences. The NodeMCU processes this information for vehicle tracking and monitoring. Overall, the circuit integrates sensors, power supply, relay control, and communication modules to detect unauthorized movement, stop the engine, and track the vehicle location effectively.

CODING AND TESTING ANALYSIS

8.1 Code Description:-

The program is written for the NodeMCU ESP8266 microcontroller to implement an IoT-based anti-theft alert system. It uses Wi-Fi connectivity, sensors, an LCD display, and a web server to monitor and display the status of a bike security system. At the beginning of the code, several libraries such as ESP8266WiFi, WiFiClient, LiquidCrystal_I2C, ESP8266WebServer, and Wire are included. These libraries provide the required functionality for Wi-Fi communication, web server creation, and interfacing with the LCD display through the I2C protocol. The Wi-Fi network credentials (SSID and password) are defined so that the NodeMCU can connect to a wireless network. Next, an LCD object is created using the I2C address 0x27 for a 16×2 LCD display, and a web server object is initialized on port 80, which is the standard port for HTTP communication. Several variables are also declared to store sensor readings and status messages. These variables are later used to display information on the LCD as well as on a web page. In the setup() function, the input and output pins of the NodeMCU are configured. Pins D6 and D7 are set as input pins to receive signals from sensors such as vibration or tampering detection sensors, while D0 is configured as an output pin to control an alert device such as a buzzer or relay. The LCD is initialized and displays a welcome message to indicate that the system has started successfully. After that, serial communication is started for debugging purposes. The NodeMCU then attempts to connect to the Wi-Fi network using the provided SSID and password. During the connection process, a “Connecting...” message is displayed on the LCD and progress is printed on the serial monitor. Once the connection is successfully established, the device prints the assigned IP address on both the serial monitor and the LCD display. This IP address can be used by the user to access the monitoring webpage from any device connected to the same network. The program also defines a web server route using `server.on("/")`.

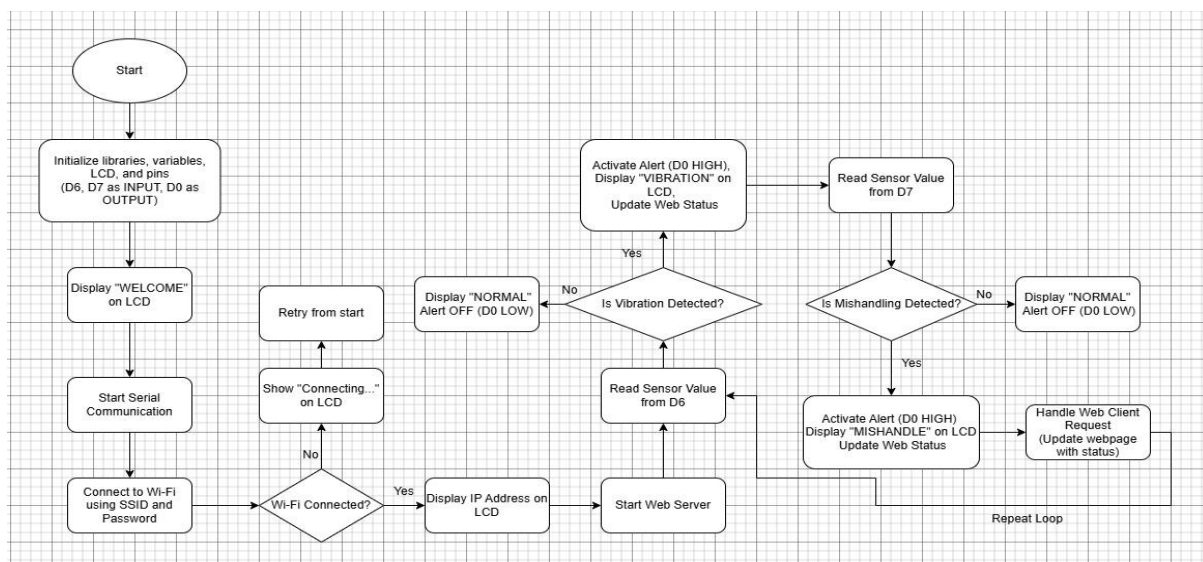


Fig 8.1.1:- Block Diagram of the Code

When a user opens the NodeMCU IP address in a web browser, the server generates a simple HTML webpage displaying the current bike status and vibration status. The webpage automatically refreshes every few seconds so that the user can continuously monitor the

system status in real time. In the loop() function, the NodeMCU continuously reads the sensor values from pins D6 and D7 using the digitalRead() function.

```

1 #include <ESP8266WiFi.h>
2 #include <WiFiClient.h>
3 #include <LiquidCrystal_I2C.h>
4 #include <ESP8266WebServer.h>
5 #include "Wire.h"
6 // Replace with your network credentials
7 const char* ssid = "admin";
8 const char* password = "123456789";
9 LiquidCrystal_I2C lcd(0x27,16,2);
10 ESP8266WebServer server(80); //instantiate server at port 80 (http port)
11
12 String page = "";
13 int data1;
14 String data2;
15 int data3;
16 String data4;
17 void setup(void){
18
19     pinMode(D6, INPUT);
20     pinMode(D7, INPUT);
21     pinMode(D0, OUTPUT);
22     lcd.begin();
23     lcd.backlight();
24     lcd.setCursor(0, 0);
25     lcd.print("      WELCOME      ");
26     delay(2000);
27     lcd.clear();
28
29     delay(1000);
30     Serial.begin(115200);
31     WiFi.begin(ssid, password); //begin Wi-Fi connection
32     Serial.println("");
33
34     // Wait for connection
35     while (WiFi.status() != WL_CONNECTED) {
36         delay(500);
37         Serial.print(".");
38         lcd.setCursor(0, 0);
39         page+="

### 


```

Fig 8.1.2:- Source code of the NodeMCU ESP8266 for the System.

If a vibration is detected (logic LOW from the sensor), the system updates the status message to indicate “Vibration Detected” and activates the alert output on pin D0. At the same time, the LCD displays a warning message. If no vibration is detected, the system shows a “Normal” status and turns off the alert output. Similarly, the second sensor connected to pin D7 checks for mishandling or unauthorized movement of the bike. If mishandling is detected, the system activates the alert and displays a warning message on the LCD and the webpage. If the condition is normal, the system continues monitoring without triggering the alert. Finally, the server.handleClient() function ensures that the NodeMCU continuously listens

for incoming web requests and updates the webpage with the latest sensor data. This allows the system to provide real-time monitoring and alerts through both the LCD display and the web interface, making it an effective IoT-based anti-theft security system.

8.2 Testing Analysis:-

The testing of the developed IoT-based anti-theft alert system was mainly carried out through functional verification of the software and hardware modules integrated with the NodeMCU ESP8266. After uploading the program to the microcontroller using the Arduino IDE, the system initialization process was observed through the Serial Monitor and LCD display. The device successfully connected to the configured Wi-Fi network using the provided SSID and password, and the assigned IP address was displayed on the LCD screen. This confirmed that the Wi-Fi communication and network configuration were working correctly. The input sensors connected to pins D6 and D7 were monitored to verify whether the program correctly detected changes in their digital states. When a disturbance signal was received from the vibration sensor, the system triggered the alert output connected to pin D0 and displayed a warning message on the LCD. Similarly, when mishandling was detected, the program updated the system status and activated the alert mechanism. These observations confirmed that the conditional statements and sensor input handling in the program were functioning properly. The web server feature of the ESP8266 was also verified by accessing the device IP address through a web browser. The generated webpage displayed the real-time status of the bike and vibration condition. The webpage refreshed periodically, ensuring that updated system information was visible to the user. This confirmed that the ESP8266 web server was successfully transmitting data over the Wi-Fi network. Based on these observations, the coding structure and logical implementation were found to be functioning as intended, enabling the system to monitor sensor inputs, trigger alerts, and provide real-time status updates through both the LCD display and the web interface.

RESULTS ANALYSIS

The results of the IoT-based Anti-Theft Alert System were analyzed by observing the responses of the LCD display, buzzer, and web interface under different conditions of sensor inputs. The system was designed to monitor vibration and mishandling of the vehicle using sensors connected to the NodeMCU ESP8266. Whenever any abnormal activity is detected, the system updates the status on the LCD display, triggers the buzzer alert, and also sends the updated status to the web page through the built-in web server. Four different use cases were tested to analyze how the system responds to different situations. The outputs obtained from the system confirm that the program logic and sensor integration work correctly and provide real-time alerts for possible theft or tampering attempts.

9.1 Normal Condition:-



Modern Security System with Anti-Theft Feature

BIKE Status:

NORMAL

Vibration Status:

NORMAL

Fig 9.1:- LCD and Webpage Output of Normal Condition

In the normal condition, both the vibration sensor and mishandling sensor remain inactive. The LCD screen displays the message “NORMAL”, indicating that no suspicious activity is detected. Similarly, the webpage accessed through the NodeMCU IP address also shows the status as NORMAL. In this situation, the buzzer remains turned off since there is no disturbance or movement detected by the system. This confirms that the system can correctly identify a safe and stable condition without triggering unnecessary alerts.

9.2 Mishandling Detected:-



Modern Security System with Anti-Theft Feature

BIKE Status:

Mishandling Detected - Lat-20.2 N Long- 85.8 E

Vibration Status:

NORMAL

Fig 9.2:- LCD and Webpage Output of Mishandling Condition

When the mishandling sensor detects unauthorized movement or tampering of the bike, the system changes its status. The LCD display shows “MISHANDLE”, indicating that the vehicle is being improperly handled. At the same time, the webpage updates the status to MISHANDLE so that the user can monitor the activity remotely. In this case, the buzzer produces a short beep sound to alert nearby individuals that suspicious movement has been detected.

9.3 Vibration Detected:-



Modern Security System with Anti-Theft Feature

BIKE Status:

NORMAL

Vibration Status:

Vibration Detected - Lat-20.2 N Long- 85.8 E

Fig 9.3:- LCD and Webpage Output of Vibration Condition

When vibration is detected by the vibration sensor, the system interprets it as a potential disturbance or attempted theft. The LCD screen displays “VIBRATION”, and the same status is also updated on the webpage interface. Similar to the mishandling case, the buzzer produces a short beep sound to warn about possible tampering. This demonstrates that the system can successfully detect vibration-based disturbances and generate an alert.

9.4 Mishandling and Vibration Detected:-

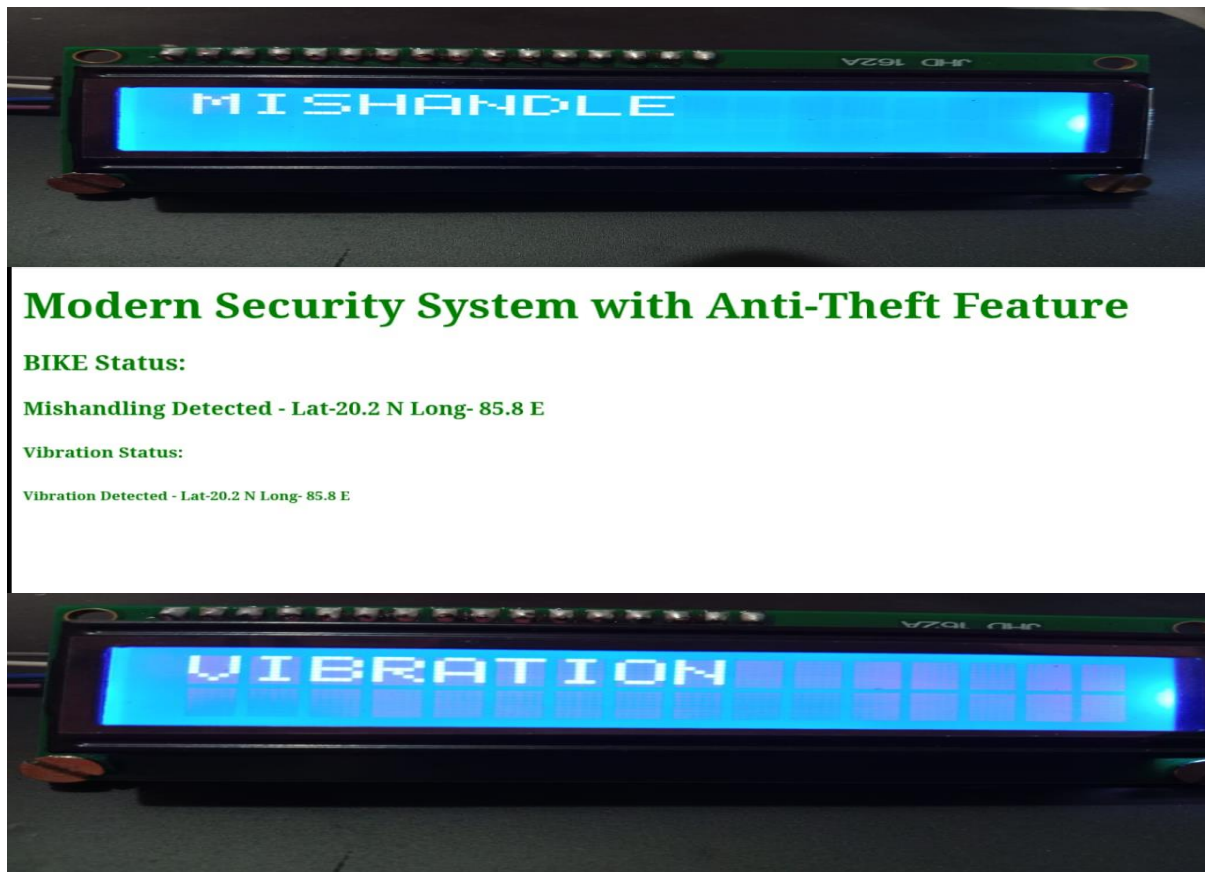


Fig 9.4:- LCD and Webpage Output of Both Conditions

In the most critical situation, when both vibration and mishandling are detected simultaneously, the system generates a stronger alert response. The LCD display indicates that both conditions are present, and the webpage also shows the combined alert status. In this case, the buzzer produces a loud continuous sound, indicating a high-risk situation where the vehicle may be under theft attempt. This condition ensures that immediate attention is drawn to the vehicle.

From the above observations, it can be concluded that the IoT-based Anti-Theft Alert System functions effectively under different operating conditions. The system successfully detects vibration and mishandling events and provides real-time alerts through the LCD display, buzzer, and web interface. The different alert levels help in distinguishing between minor disturbances and critical theft attempts. Overall, the obtained results demonstrate that the developed system is capable of monitoring vehicle safety and providing timely alerts to prevent theft.

ADVANTAGES AND LIMITATIONS

10.1 Advantages:-

- **Dual-Sensor Verification:** By combining a Joystick and a Vibration/Tilt sensor, the system cross-checks data to ensure a theft is actually happening before sounding the alarm.
- **Active Immobilization:** Unlike passive trackers, this system uses a Relay to physically cut the ignition, making the vehicle impossible to start.
- **Global Connectivity:** Using IoT (NodeMCU), the owner receives real-time alerts on their smartphone regardless of the distance from the vehicle.
- **Live GPS Tracking:** The integration of a GPS module allows the owner to see the exact real-time coordinates of the vehicle on a map for quick recovery.
- **Cost-Effective Solution:** Built with affordable, open-source components, making high-end security available for a fraction of the price of commercial systems.
- **Local Visual Feedback:** The 16x2 LCD provides the user with immediate status updates during routine operation.
- **Anti-False Trigger Logic:** The software uses threshold-based sensing, so the alarm isn't triggered by wind, heavy rain, or passing trucks.
- **Scalability:** The system can be easily upgraded with more sensors (like an ultrasonic or tilt sensor) without changing the core hardware.
- **Visual and Audio Deterrents:** The combination of a high-decibel Buzzer and a flashing LCD alert acts as a strong psychological deterrent for thieves.
- **Low Latency:** Using Wi-Fi protocols ensures that the notification reaches the owner within seconds of the breach detection.

10.2 Limitations:-

- **Internet Dependency:** The IoT features and mobile alerts require a stable Wi-Fi connection; without internet, the system only works as a local alarm.
- **GPS Cold Start:** The GPS module may take a few minutes to lock onto satellites when the vehicle is first turned on or moved from a garage. In the current prototype phase, GPS coordinates are simulated via serial data packets to demonstrate the integration of location-based alerts within the web dashboard.
- **Power Drainage:** Continuous monitoring by the NodeMCU and LCD can slowly drain the vehicle's battery if the bike is left parked for several weeks.
- **Signal Obstruction:** GPS and Wi-Fi signals can be blocked or weakened if the vehicle is parked in an underground basement or inside a metal container.
- **Hardware Fragility:** As a prototype, the breadboard or jumper wire connections can be sensitive to the high vibrations of a real running engine.
- **Environmental Sensitivity:** High humidity or extreme heat could potentially affect the sensitivity of the analog joystick over long periods.
- **Manual Arming:** In the current version, the user must remember to arm the system via the app; it does not yet feature "Auto-Arming" when the key is removed.
- **Single Point of Failure:** If the NodeMCU loses power or crashes, the entire security system (including the engine lock) might be bypassed or disabled.

FUTURE SCOPE

Future improvements of the system could include several advanced features to enhance functionality, security, and reliability. Integrating a GSM module would allow the system to send SMS alerts and location data even in areas without Wi-Fi coverage. The manual “Arm/Disarm” feature could be replaced with biometric authentication such as a fingerprint sensor or Face ID using a camera module for keyless ignition. A dedicated mobile application for Android or iOS could also be developed with a built-in map to track the vehicle’s movement history and speed. To improve power reliability, a small solar panel could be added to trickle-charge the backup battery, ensuring the system remains active even if the main vehicle battery is disconnected. Security accuracy could be enhanced using AI-based motion detection, where an accelerometer combined with machine learning distinguishes between normal contact (someone sitting on the bike) and suspicious activity such as lock tampering. The system could also include voice alerts using a text-to-speech module to warn intruders with messages like “Unauthorized access detected, owner notified.”

Instead of only sending live alerts, cloud data logging using services like Firebase could store system data, allowing the owner to view a history log of movements or tampering events over the last 30 days. Another improvement could be geofencing, where a safe zone such as a home or college location is defined; if the GPS detects the bike moving outside a 50-meter boundary without disarming the system, the engine would automatically lock and the owner would be notified. To reduce battery consumption, smart power saving (deep sleep mode) could be implemented so that the NodeMCU wakes up only when the vibration or tilt sensor detects movement. An emergency SOS button could also be hidden on the bike, enabling the owner to secretly send a help message with their live location to family or authorities during a robbery. For real-vehicle integration, the relay could be connected to the fuel pump for remote fuel cut-off, causing the bike to stall shortly after being started by an unauthorized person. The system could also be connected to smart wearables, allowing alerts to appear instantly on a smartwatch through vibration notifications. Finally, using a weatherproof IP67 enclosure made through professional 3D printing or industrial plastic would protect the NodeMCU and sensors from rain, dust, and engine heat, making the system suitable for long-term real-world road conditions.

CONCLUSION

The development and successful implementation of the **IoT Based Smart Anti-Theft Alert and Engine Lock System** mark a significant departure from traditional, passive security measures to a more intelligent, active ecosystem. As established in the initial problem statement, the primary failure of modern two-wheeler security lies in the "passive gap" the inability of a vehicle to defend itself or communicate once a physical lock is bypassed. This project has effectively addressed that gap by transforming a static vehicle into an intelligent node within the Internet of Things. By shifting the detection logic from simple vibration sensing to active steering orientation monitoring via a Joystick sensor, the system has proven its ability to solve the industry-wide problem of high false-alarm rates. Through experimental testing on the prototype, it was observed that the logic successfully detects any sustained movements like handle-turning or rolling as well as vibration. Furthermore, the project successfully demonstrated the power of Real-Time Immobilization. In a conventional scenario, a thief can operate in silence or ignore an alarm, however, this system's ability to act via the Relay-driven engine lock provides a physical barrier that renders the vehicle useless to an intruder. The transition of this logic from a digital signal to the physical stopping of a DC motor provides a clear, verifiable proof-of-concept for how this would function on a real-world ignition coil or fuel pump. The integration of the NodeMCU (ESP8266) ensures that the Distance Barrier is permanently removed. By utilizing IoT protocols, the system successfully bridged the physical vehicle to the owner's smartphone, providing instantaneous push notifications and GPS coordinates.

This ensures that the owner is empowered with actionable data rather than just an acoustic signal that might go unheard. From an academic and economic standpoint, this project demonstrates that high-end, intelligent security does not necessarily require expensive proprietary hardware. By utilizing affordable, open-source components and optimizing them through smart software logic, we have created a scalable model that is accessible to the average consumer. The dual-layered verification approach using both the joystick and the vibration/tilt sensor ensures a high degree of reliability that minimizes alarm fatigue. In conclusion, this prototype successfully validates the integration of embedded systems, sensor fusion, and wireless communication. It stands as a robust foundation for future automotive security, proving that digital connectivity and active immobilization are the keys to safeguarding millions of two-wheelers in the modern era. While currently a functional experimental model, the results clearly indicate that this architecture is a viable, reliable, and necessary evolution in the fight against vehicle theft.

REFERENCES

1. Mallieswaran, K., et al. *Design and Implementation of Advanced Safety Features in Two-Wheeled Vehicles for Accident Prevention and Anti-Theft System*. SAE Technical Paper, 2024..
2. Gnanapriya, S., et al. "An IoT based anti theft detection and notification system for two wheelers." *Proceedings of the 7th International Conference on Innovations and Research in Technology and Engineering (ICIRTE-2022)*, organized by VPPCOE & VA, Mumbai-22, INDIA. 2022.
3. Karthik, G. Yogindra Siva Sai, et al. "Anti-Theft and Engine Locking System using GSM Technology." *2024 International Conference on Augmented Reality, Intelligent Systems, and Industrial Automation (ARIIA)*. IEEE, 2024.
4. Poorvaja, V. S., et al. "Smart IoT-based System with Enhanced Safety and Accident Prevention for Two-Wheelers." *2025 6th International Conference on Inventive Research in Computing Applications (ICIRCA)*. IEEE, 2025..
5. Ramprakaash, K., et al. "IoT based Tracking System for Two Wheelers." *i-Manager's Journal on Embedded Systems* 9.2 (2021): 15.
6. Abdullahi, I. M., et al. "Development of a Smart Internet of Things Based Motorcycle Theft Prevention System." *ARID ZONE JOURNAL OF ENGINEERING, TECHNOLOGY AND ENVIRONMENT* 21.3 (2025): 847-856.
7. Rakhmadi, Aris, et al. "An IoT-Based Motorcycle Security System Using Arduino and Android for Theft Prevention." *Compiler* 14.1 (2025): 43-52.
8. Paul, Rinika, K. Deepa, and M. Nithya. "Implementation of airbags and Iot-based safety and alert systems in two-wheelers." *2022 IEEE 19th India Council International Conference (INDICON)*. IEEE, 2022.
9. Prakash, K. Jeya, et al. "Vehicle theft intimation over sms and remote control of its engine." *2021 6th International Conference on Communication and Electronics Systems (ICCES)*. IEEE, 2021.